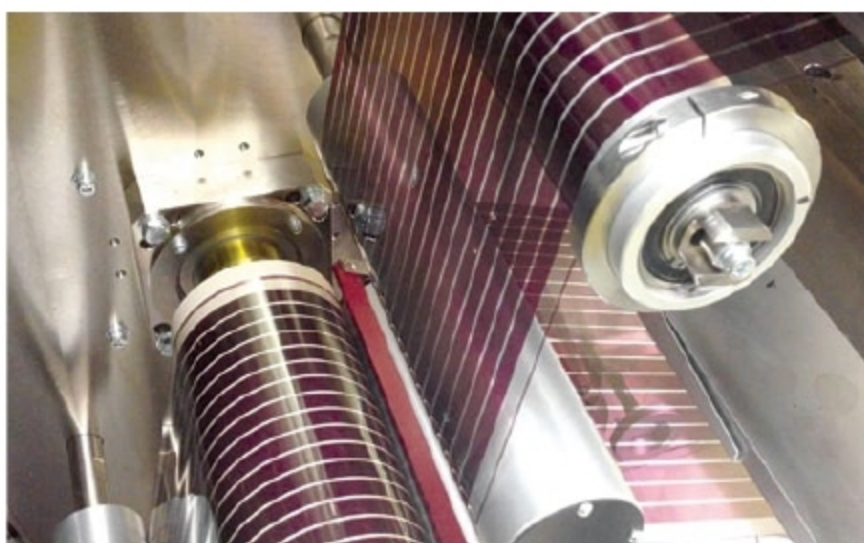


Towards A Sustainable Future Driven By The Thrust of Organic, Flexible and Printable Solar Power

► The global organic solar cells market is expected to generate about US\$ 290 million by 2027 at a CAGR of 21 per cent

Do you know that we are consuming natural resources at a much faster pace than we can possibly imagine? And that buildings make up for about forty per cent of the total demand for energy? Also, the fact that our buildings can be made in such a way that they are completely sustainable on their own?

Organic solar cells have arrived and how! These are thin, flexible and printable. Organic solar photovoltaic



Organic solar cells being printed (Credit: www.businesskorea.co.kr)

cells can be printed roll-to-roll just like newspapers and be ready for use. Unlike traditional silicon solar cells, printed solar cells are made using organic semiconducting polymers. These can be dissolved in a solvent and used with ink.

Demand for organic photovoltaics is increasing in India. Gurpreet Singh Tikku, founder - Mister Tikku, shares, "I saw some beautiful canopies near Miracle Garden during my recent visit to Dubai. These were magnificent in purple and grey during the day and colourfully lit up at night. I questioned the people around, out of curiosity, whether these have a purpose to solve. To my surprise, I was told that apart from providing shade from the Sun's scorching heat, these were induced with solar panels, made of organic cells, to light up the area. As my trip continued, I found these were prevalent in several housing complexes and restaurants."

He adds, "If these were available in India, I would like to explore these for my office and home."

Karan Sharma, director - SunSmith Infra, says, "I have been into the business of solar panels for almost a decade now. In the past couple of years demand for organic solar cells has definitely risen. Most of our business comes from the implementation of traditional rooftop solar panels, mostly in the enterprise sector.

"However, of late, there has been demand to implement customised organic solar panels in residential areas, too. We have been reaching out to manufacturers to fulfil such demands. So far, it is only need based."

Business growth drivers

The global organic solar cells market is expected to generate about US\$ 290 million by 2027 at a CAGR of 21 per cent. This is being regarded as a highly lucrative future business area.

Main factors contributing to the growth of organic solar cells are the bend of electronics towards utilising renewable sources of energy and immense support from the government and other prominent national and international institutions. Better manufacturing technologies for polymer solar cells are also being developed. The cells are lightweight and can be cost-optimised based on the materials being used to manufacture these. Polymers have great light absorption power and offer minimal loss of energy.

Growth challenges

Apart from the many application benefits provided by organic solar cells, one major limiting factor towards its mass usage is that these come with limited operational life. These may be more efficient than traditional solar cells, but their functional life span is limited to two decades only. Research around the world is going on to develop polymer solar cells combined with various materials to increase their shelf life.

The scale of applications for organic, printable and flexible solar cells is huge, from being the best option for tiny mobile devices to providing great appeal for tall standing structures. The polymer solar cell market is bound to grow and expand. Related businesses are also expected to flourish in the future. **EFY**

—Nidhi Arora, executive editor, EFY



Organic solar cells projected growth rate (Credit: www.futuremarketinsights.com)